

Subject: Cloud Computing

• Section-A (5x2=10)

1. What is fault tolerance?

Ans Fault tolerance refers to the system's ability to continue operating smoothly even if one or more components fail. It ensures high ~~ability~~ availability and reliability by using redundancy, load balancing and failover mechanisms.

2. What is virtualization?

Ans Virtualization is a technology that creates virtual versions of physical hardware, like a server, network, or storage device. It uses software to simulate hardware functionality.

3. What is Load Balancing?

Ans Load balancing is the process of ~~the~~ creating virtual instances of computing resources.

Load balancing is the technique of distributing network traffic or workloads across multiple servers to ensure no single server is overwhelmed, improving performance and reliability.

4. Define Scalability.

Ans Scalability refers to a system's ability to handle increasing workloads by adding resources (scale-up or scale-out) while maintaining performance and efficiency.

5. State Comparative analysis of requirement.

Ans A comparative analysis of requirements involves evaluating different cloud solutions based on factors like cost, performance, security, scalability and compatibility to determine the best fit for a specific use case.

Subject: _____

Date: ____/____/____

• Sec-B (3x7=21)

6. Explain Data Management Storage and Processing.

Ans Data Management, Storage and Processing are essential components of modern IT infrastructure. They enable organizations to efficiently collect, store and analyze data while ensuring security, accessibility and compliance with regulatory requirements.

(i) Data Management - Data Management involves organizing, storing and maintaining data efficiently to ensure security, integrity and accessibility. It includes database management, data governance and backup strategies.

(ii) Data Storage - Data Storage refers to saving digital information using various methods such as cloud storage, database or physical storage devices. - Cloud providers offer storage solutions like object storage (Amazon S3), block storage and file storage for efficient data retrieval and management.

(iii) Data Processing: - Data processing involves transforming raw data into meaningful insights. It includes batch processing (processing large amounts of data in batches) and real-time processing (analyzing data instantly as it arrives). Cloud platforms use big data technologies like Hadoop, Spark and Serverless Computing for scalable data processing.

7. Explain Virtualization Management?

Ans Virtualization management refers to the administration and control of virtualized resources, such as servers, storages and networks to optimize performance, scalability and efficiency in cloud environment.

Key Components:-

- Virtual Machine Management - creating, deploying and monitoring virtual machines.

Subject: _____

- Resource Allocation - Distributing CPU, Memory and storage efficiently.
- Load Balancing - Ensuring optimal workload distribution across virtual resources.
- Security & Compliance - Implementing access controls and compliance policies.
- Backup & Recovery :- Protecting data through regular snapshots and backups.

Tools & Technologies :-

Popular virtualization management tool include - VMware vSphere, Microsoft Hyper-V, KVM, and Citrix XenServer. Cloud providers like AWS, Azure and Google cloud offer built-in virtualization management solutions.

8. State Scalability and Load Balancing

Ans Scalability :- it refers to a system's ability to handle increasing workloads by adding resources. it can be of two types.

- Vertical Scaling (Scale-up) - Increasing the capacity of existing resources (e.g. - upgrading a server's CPU/RAM).
- Horizontal Scaling (Scale-out) :- Adding more instances of resources (e.g. - multiple servers handling requests).

Load Balancing - it is the process of distributing network traffic or workloads across multiple servers to prevent overload and ensure efficient resource utilization. It improves system performance, reliability and availability.

Overall, Scalability and load balancing work together to enhance system performance and ensure seamless cloud computing operations.

Subject: _____

Date: ____/____/____

• Sec - C (3x11 = 33)

9. Explain cloud Deployment for Enterprises.

Ans Cloud deployment refers to how cloud computing services are implemented and made available to enterprises. It involves selecting the right deployment model based on business needs, security and scalability requirements.

- Deployment Models:- Enterprises can choose from different cloud deployment models:-

- Public Cloud - Services are hosted by third-party providers like AWS, Azure and Google cloud, offering scalability and cost effectiveness.
 - Private Cloud - A dedicated cloud infrastructure for a single organization, ensuring high security and control.
 - Hybrid Cloud - A combination of public and private clouds, allowing flexibility and optimization resource utilization.
 - Community cloud - A shared infrastructure for organization with common requirements, such as government agencies.
- Service Model →
- IaaS (Infrastructure as a Service) - Provide access to fundamental computing resources like ~~service~~ servers, storage, and networks.
 - PaaS (Platform as a Service) - Offers a platform for developing, deploying and managing applications, including tools and infrastructure.
 - SaaS (Software as a Service) - Delivers software applications over the internet, allowing users to access them without installing or managing software.

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Subject: _____

- Benefits for Enterprises: -
 - Cost Benefits - Reduces IT infrastructure costs by using Pay-as-you-go models.
 - Scalability - Easily adapts to growing business demands.
 - Security & compliance - Private and hybrid models offer enhanced security and regulatory compliance.
 - Business Continuity - Ensures data backup and disaster recovery capabilities.
 - Improved Collaboration - Enables remote work and global access to resources.

10. Explain User Requirement.

Ans User Requirement are the specific needs, expectations and desires of the end-users for a product, system or service, which are crucial for ensuring that the final product meets user needs and expectations.

- Types of User Requirements: -

- Functional Requirements - These specify what the system should do. ex: "The system allow user to log in with a Username and Password".
- Non-Functional Requirements - These define how the system performs. ex:- "The system should load the dashboard within 2 seconds".
- User Interface Requirements: - Defines how the user interacts with the system.
Ex: "The interface should be mobile-responsive and easy to navigate".

Subject: _____

Date: / /

- Importance —

- (i) Increased User Satisfaction - By understanding and fulfilling user requirements, developers can create products that users find easy to use that meet their specific needs.
- (ii) Helps in designing systems that meet actual user needs.
- (iii) Provide a clear scope for developers and stakeholders.
- (iv) Prevents miscomm. and costly revision later in project.

- Methods to Gather User Requirements: -

- Interviews and surveys
- Focus groups
- Observation and field studies
- Use case and persona creation.

11. Explain Fault tolerance.

Ans Fault tolerance is the ability of a system to continue operating properly in the event of the failure of one or more of its components. In cloud computing it ensures high availability, reliability and continuous service even when parts of the system fail.

- Importance of Fault Tolerance: -

- (i) Business continuity - Ensures that critical applications and systems remain operational even in the face of failures, minimizing downtime and potential losses.
- (ii) High Availability - Reduces the risk of service disruptions and ensures that users can access applications and systems when they need them.
- (iii) Data Integrity - Protects data from loss or corruption by

Subject: _____

replicating it across multiple locations and implementing mechanisms for recovery.

- User Experience - Minimize disruptions to users and ensure that they can continue to work without experiencing any noticeable downtime or performance issues.

• Techniques Used for Fault Tolerance :-

- (i) Redundancy - Multiple instances of components (e.g. servers, storage) are used so that if one fails, others take over.
- (ii) Failover Systems :- Automatically switch to a standby system or server when the primary one fails.
- (iii) Load ~~Stat~~ Balancing - Distributes workloads evenly to avoid overloading and reduce the chance of failure.
- (iv) Backup or snapshots - Regular backups ensure that data can be restored after a failure.
- (v) Monitoring and Alerts - Systems are continuously monitored for failures, and alerts are generated to take immediate action.

• Cloud Examples:

- (i) AWS Availability Zones - Replicate data and services across multiple zones.
- (ii) Google Cloud's Live Migration - Keeps VMs running even during maintenance.
- (iii) Azure Site Recovery - Provides disaster recovery services.